Sprint 4 Retrospektive
Projekt: KI-gestütztes System zur frühzeitigen Erkennung von Datenlecks
Sprint: Sprint 4

The primary goal of this sprint was to bring together previously developed components to create a functional and presentable final product. The overall objective of the project was to develop an AI-powered system capable of early detection of potential data breaches involving selected companies on dark web sources. The target user group consisted primarily of security analysts, corporate IT security teams, and organizations seeking to identify data breach risks at an early stage. Therefore, it was important for the product not only to collect data but also to analyze and classify it in an understandable way and make it evaluable via a dashboard.

Throughout the project, the team worked with a generally strong sense of teamwork. By the end of Sprint 4, the project had reached a state where all core components were integrated, functional, and ready for presentation. The Backend, Collector, AI Analysis, and Frontend work together to demonstrate the process of data collection, analysis, and evaluation via the dashboard. While some automation goals could not be fully implemented due to technical limitations and time constraints, the system's core concept and basic product flow were successfully demonstrated.

By the end of Sprint 4, many features were either fully completed or functional at the demo level. On the Collector side, a data collection flow was established from the selected sources, and the basic technical infrastructure for accessing these sources via a Tor connection was put in place. The process of storing the collected findings in the database and transferring them from the backend to the frontend dashboard was implemented. The backend APIs have been integrated with the dashboard and can now provide the necessary data for summary cards, findings, reports, charts, and source management. On the frontend, features such as the findings table, filtering, sorting, and pagination have been developed. Additionally, the company matching logic was improved, and the number of findings remaining as "Unknown" was significantly reduced. LLM analysis is no longer fully automated but can now be initiated manually, and the ability to save analysis results was implemented. The Reports section was made more user friendly, and overall, the system has reached a functional product flow ready for presentation at the final demo.

Nevertheless, some of the initially planned items could not be fully implemented or were removed from the project scope. Certain security and legal documents that were intended to be prepared by the end of the project were excluded from the scope. The process of collecting data from dark web sources could not be fully stabilized; obstacles such as Tor connection issues, access problems, and CAPTCHAs occurred intermittently throughout the project. Since the LLM analysis could not be run fully automatically, the manual "Analyze with AI" approach was chosen as a more suitable solution. Some reporting and export features were implemented on the front end, but the full-scale backend export system remained limited. Although the company matching logic was improved, a 100% accurate match could not be guaranteed for every finding.

From a technical perspective, one of the most successful aspects of Sprint 4 was the significant progress made in frontend and backend integration. The dashboard now displays more realistic, consistent, and usable data. Manually triggering and recording LLM results proved to be a practical and reliable solution within the project's constraints. Improvements to the company matching logic enhanced data quality and helped the system generate more meaningful results from the analyst's perspective. Additionally, the clutter that had developed on the frontend in previous sprints was transformed into a cleaner and more reliable user experience by the end of Sprint 4. As a result, the project was brought to a suitable level for both individual presentations and the final presentation.

Looking back, if the project had been restarted from the very beginning, certain aspects could have been handled differently. In particular, it would have been beneficial to define role assignments more clearly from the start. As the project progressed, it became apparent that some issues and tasks were better suited to the areas of expertise of other team members. While this increased collaboration within the team, it also led to time losses on certain tasks. Similarly, defining issues and milestones more clearly from the start could have streamlined the process. During the project, prioritizing some issues, postponing others, and the emergence of new requirements toward the end occasionally created complexity and uncertainty.

One of the areas where we struggled the most as a team was front-end design. Since each team member had different expectations and suggestions, it took time to reach a consensus. Looking back, it was the right decision to prepare two different frontend designs during Sprint 4 and make the final decision after discussing it with the Betreuer. Thanks to this decision, trust within the team grew, and a design that everyone could accept was selected for the final presentation. Development efforts were then focused on this design. Although this process took some time initially, it was an important and correct decision in terms of the project's final appearance.

In terms of the process, the team was generally well organized. When one team member got stuck at a certain point, another team member would usually step in to help, and the problem was solved together. Maintaining constant communication and consulting with one another as needed while working on issues throughout the project provided a significant advantage. One of the most effective methods was performing merge operations in a sequential and logical order. In the beginning, since everyone merged their changes whenever they wanted, code conflicts and merge issues arose. Later, when the merge order was planned more carefully, these problems decreased, and the team saved time.

The team also held a separate meeting to prepare for the final presentation, the poster, and individual presentations. During this meeting, the team emphasized that the poster should not consist solely of long blocks of text, but should be visually engaging and able to quickly convey the project's main idea. A basic structure for the poster was established, and a new meeting was scheduled to finalize the design. As for the individual presentations, the most important point is for each team member to clearly explain their own responsibilities throughout the project, the problems they encountered, the decisions they made, and the outcomes of those decisions. For this reason, the presentations should not only showcase the final product but also highlight the development process, technical challenges, lessons learned, and the team's decision-making processes.

Overall, Sprint 4 was successful in terms of completing the project and preparing it for presentation. By the end of the project, the system's core concept became clearly demonstrable: collecting and analyzing potential data leaks, matching them with companies, and making them evaluable by analysts via a dashboard. For this reason, Sprint 4 was not only the final development sprint but also one of the most important sprints during which the project was technically consolidated, team decisions were clarified, and preparations for the final presentation were made.